

Why $A \oplus B$ is a Biproduct

by notes on Freyd & Scedrov, Categories · Allegories

§1.591 — finite products and coproducts coincide in an abelian category

In any **bicartesian** category a map from a coproduct to a product is a 2×2 matrix of hom-set elements. In an **abelian** category the canonical comparison map $u : A + B \rightarrow A \times B$ becomes an **isomorphism**. Its inverse is **not** a matrix — it must be built from the addition on hom-sets, $v = \pi_A \iota_A + \pi_B \iota_B$. We construct v , verify $uv = 1$ and $vu = 1$, and watch the whole thing degenerate in SET.

Remark 0.1 (Conventions (Freyd)).

Coproduct injections $\iota_A : A \rightarrow A + B$, $\iota_B : B \rightarrow A + B$; product projections $\pi_A : A \times B \rightarrow A$, $\pi_B : A \times B \rightarrow B$. The hom-set $(A, B) = \{\text{morphisms } A \rightarrow B\}$.

§1. The 2×2 matrix

A morphism **out of a coproduct** is one map per summand; a morphism **into a product** is one map per factor. Stacking the two universal properties, a single morphism is determined by four free morphisms.

Definition 1.1 (Matrix of a map $\text{sum} \rightarrow \text{product}$).

In any bicartesian category a morphism $f : A_1 + A_2 \rightarrow B_1 \times B_2$ is **uniquely** described by

$$f \text{ is } \begin{pmatrix} f_{11} & f_{12} \\ f_{21} & f_{22} \end{pmatrix}, \quad f_{ij} = \iota_i f \pi_j \in (A_i, B_j).$$

The four entries are **unconstrained** hom-set elements. This much holds in SET too.

§2. The canonical comparison map

Specialise to $A_1 = B_1 = A$ and $A_2 = B_2 = B$. The **canonical map** $u : A + B \rightarrow A \times B$ is the one carrying the identity pattern:

$$u \text{ is } \begin{pmatrix} 1_A & 0 \\ 0 & 1_B \end{pmatrix},$$

that is, the four equations

$$\begin{aligned} \iota_A u \pi_A &= 1_A, & \iota_A u \pi_B &= 0, \\ \iota_B u \pi_A &= 0, & \iota_B u \pi_B &= 1_B. \end{aligned}$$

△ Zero morphism

The off-diagonal **0** is the **zero morphism** — the unique map through the zero object, $0_{A,B} : A \rightarrow 0 \rightarrow B$.

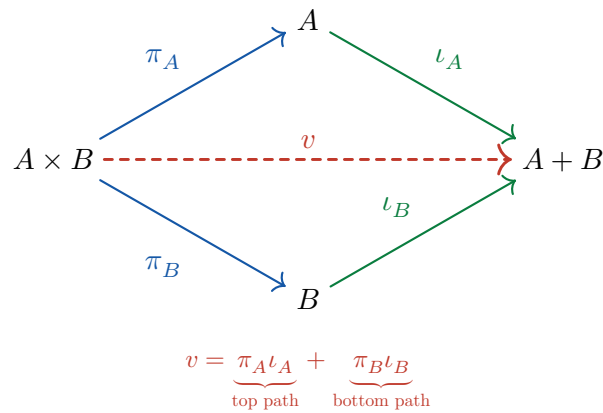
§3. The inverse is not a matrix

The matrix trick described maps **sum** $\rightarrow$ **product**. The inverse runs the other way, $v : A \times B \rightarrow A + B$ — **out of a product, into a sum** — and **neither** direction has a componentwise universal property. So v cannot be read off a matrix; it must be **built**, using the **addition on hom-sets** that an abelian category provides.

Definition 3.1 (The inverse, built with $+$).

$$v := \pi_A \iota_A + \pi_B \iota_B \in (A \times B, A + B).$$

Here $\pi_A \iota_A$ and $\pi_B \iota_B$ are two morphisms $A \times B \rightarrow A + B$, summed in the abelian group $(A \times B, A + B)$. **This step is impossible in SET** — functions don't add.



Each summand is one corner trip — out along a projection π (blue), in along an injection ι (green) — and the two trips are **added** (red) in the hom-group.

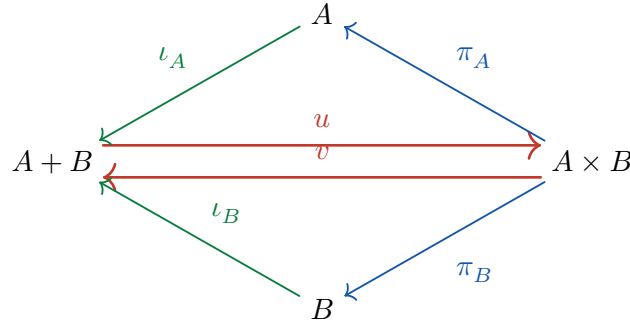
§4. u is an isomorphism

Theorem 4.1 (Biproduct).

In an abelian category the canonical map $u : A + B \rightarrow A \times B$ is an isomorphism, with inverse $v = \pi_A \iota_A + \pi_B \iota_B$. Hence finite products and coproducts coincide.

Both round trips collapse on **one** picture: the biproduct diamond carrying both u and v . Composition is bilinear, and the corner relations $\iota_i u \pi_j = 1$ if $i = j$, else 0 (the matrix of u) do all the work. We draw it twice — each time with the composite's **start** object on the left.

Proof that $uv = 1_{A+B}$ — out of $A + B$, restrict along the injections ι_A, ι_B .

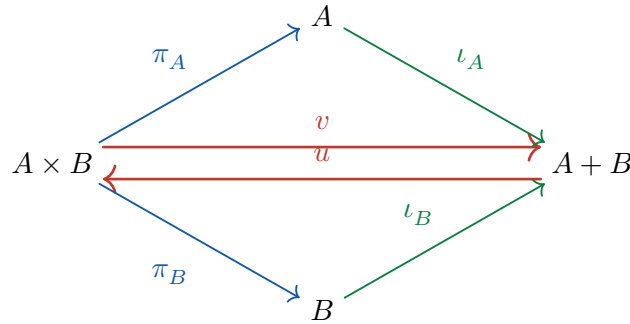


Going $\iota_A \rightarrow u$ lands in $A \times B$ with coordinates $\iota_A u \pi_A = 1_A$ (top loop) and $\iota_A u \pi_B = 0$ (crossing); then $v = \pi_A \iota_A + \pi_B \iota_B$ reassembles them:

$$\iota_A u v = (\iota_A u \pi_A) \iota_A + (\iota_A u \pi_B) \iota_B = 1_A \iota_A + 0 = \iota_A,$$

and likewise $\iota_B u v = \iota_B$. A map out of $A + B$ agreeing with 1 on both injections **is** 1.

Proof that $vu = 1_{A \times B}$ — into $A \times B$, read coordinates along π_A, π_B . The same diamond, now $A \times B$ on the left so the trip $v \rightarrow u$ starts there:



Expanding $v = \pi_A \iota_A + \pi_B \iota_B$ and using $\iota_i u \pi_j \in \{1, 0\}$:

$$vu \pi_A = \pi_A (\iota_A u \pi_A) + \pi_B (\iota_B u \pi_A) = \pi_A 1_A + \pi_B 0 = \pi_A,$$

and likewise $vu \pi_B = \pi_B$. A map into $A \times B$ agreeing with 1 on both projections **is** 1. Hence u is an isomorphism. $\square$

§5. Concrete picture in $\mathbf{Ab}$

Example.

In abelian groups both $A + B$ and $A \times B$ have the same underlying set $A \oplus B = \{(a, b)\}$, with

$$\iota_A : a \mapsto (a, 0_B), \quad \iota_B : b \mapsto (0_A, b), \quad \pi_A : (a, b) \mapsto a, \quad \pi_B : (a, b) \mapsto b.$$

The inverse **reassembles** a pair by adding its two one-sided pieces:

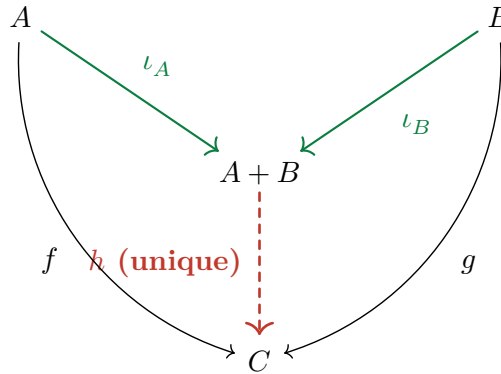
$$v(a, b) = \underbrace{(\pi_A \iota_A)(a, b)}_{\iota_A(a)} + \underbrace{(\pi_B \iota_B)(a, b)}_{\iota_B(b)} = (a, 0_B) + (0_A, b) = (a, b).$$

That single $+$ is the **group operation** of $A \oplus B$ — the abelian structure doing the work the universal properties cannot.

The map $0 \rightarrow B$ sends the one element of the trivial group $0 = \{0\}$ to B 's identity 0_B (homomorphisms preserve identities), so the zero morphism $A \rightarrow 0 \rightarrow B$ is the constant map $a \mapsto 0_B$.

§6. Why the coproduct lands on the same set

The product $A \times B = \{(a, b)\}$ is the usual Set product. The surprise is that the **coproduct** is the **same** set. By definition $A + B$ is a coproduct iff for every pair of homomorphisms $f : A \rightarrow C$, $g : B \rightarrow C$ there is a **unique** $h : A + B \rightarrow C$ with $h\iota_A = f$ and $h\iota_B = g$:



h is forced. Since $(a, b) = \iota_A(a) + \iota_B(b) = (a, 0) + (0, b)$, any such h obeys

$$h(a, b) = h\iota_A(a) + h\iota_B(b) = f(a) + g(b).$$

h is a homomorphism — but only because C is abelian. Respecting $+$ means

$$h((a, b) + (a', b')) = f(a + a') + g(b + b') = f(a) + f(a') + g(b) + g(b')$$

must equal $h(a, b) + h(a', b') = f(a) + g(b) + f(a') + g(b')$ — and it does **only** after swapping $g(b)$ past $f(a')$, i.e. using commutativity of C . In a non-abelian target this fails and the coproduct balloons into the free product $A * B$; abelianness is exactly what collapses $A + B$ back onto the product set $\{(a, b)\}$, with $\iota_A(a) = (a, 0_B)$, $\iota_B(b) = (0_A, b)$.

Theorem 6.1 (Punchline).

In SET you cannot add $(a, 0_B)$ to $(0_A, b)$, there is no zero object, and

$$|A + B| = |A| + |B| \neq |A| \cdot |B| = |A \times B|.$$

Sums and products genuinely differ. In an abelian category the hom-sets are abelian groups, the inverse $v = \pi_A \iota_A + \pi_B \iota_B$ exists, and $A + B \cong A \times B$: the **biproduct** $A \oplus B$.